Big Data

1. Importing from JSON

Exp: 1
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Study of Various Network Commands used in Linux & Windows

Aim:

To study and work various network commands used for Linux & Windows for networking configuration and diagnostics.

Commands:

1. arp (Address Resolution Protocol).
   - Displays and manages the App table, which maps IP address to MAC addresses.
2. hostname - Displays / Changes the name of local computer.
3. ipconfig - Displays and configures the IP address subnet mask, and default gateway.
4. nbstat - Displays net BIOS over TCP/IP statistics and NetBIOS name tables.
5. Netstat - Displays active network connections, rowing tables, and listing ports.
6. nslookup - Queries DNS servers to find the IP address or Domain name of a host.

7. pathping :- This command traces route to the destination adress then launch a 25 second test for each router along the way.

8. ping - It is the best way to test connectivity between two nodes.

9. route :- Route command is used to show / manipulate the IP routing table.

```
C:\Users\HDC0222053>arp -a

Interface: 172.16.10.44 --- 0x8
  Internet Address      Physical Address      Type
  172.16.8.1            7c-5a-1c-cf-be-45     dynamic
  172.16.9.209          28-a4-4a-96-01-8b     dynamic
  172.16.10.4           d8-bb-c1-c5-ca-10     dynamic
  172.16.10.5           d8-bb-c1-c5-cb-a3     dynamic
  172.16.10.7           d8-bb-c1-c5-cd-8b     dynamic
  172.16.10.8           d8-bb-c1-c5-cc-af     dynamic
  172.16.10.9           d8-bb-c1-c5-cb-6f     dynamic
  172.16.10.12          d8-bb-c1-c5-c8-a8     dynamic
  172.16.10.13          d8-bb-c1-c5-cd-5c     dynamic
  172.16.10.14          d8-bb-c1-c5-ca-4b     dynamic
```

```
C:\Users\HDC0222053>hostname
DESKTOP-NSJ07FM
```

```
C:\Users\HDC0222053>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::4d1e:9ee7:96bb:fd4e%8
   IPv4 Address. . . . . . . . . . . : 172.16.10.44
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1
```

```
C:\Windows\system32>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName    Remote host machine name.
  IP address    Dotted decimal representation of the IP address.
  interval      Redisplays selected statistics, pausing interval seconds
                between each display. Press Ctrl+C to stop redisplaying
                statistics.


C:\Windows\system32>
```

```
C:\Users\HDC0222053>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.16.10.44:7680      DESKTOP-9NNUCNH:62952  TIME_WAIT
  TCP    172.16.10.44:7680      172.16.10.93:60510     TIME_WAIT
  TCP    172.16.10.44:7680      172.16.10.194:49627    TIME_WAIT
  TCP    172.16.10.44:61870     4.213.25.242:https     ESTABLISHED
  TCP    172.16.10.44:61916     sd-in-f188:5228        ESTABLISHED
  TCP    172.16.10.44:62125     a23-44-10-67:https     CLOSE_WAIT
  TCP    172.16.10.44:62126     172.202.64.254:https   CLOSE_WAIT
  TCP    172.16.10.44:62127     150.171.110.83:https   CLOSE_WAIT
  TCP    172.16.10.44:62128     wstr9a:https           CLOSE_WAIT
  TCP    172.16.10.44:62129     204.79.197.222:https   CLOSE_WAIT
  TCP    172.16.10.44:62157     lcmaaa-am-in-f5:https  ESTABLISHED
  TCP    172.16.10.44:62158     123:http               TIME_WAIT
  TCP    172.16.10.44:62163     123:http               TIME_WAIT
  TCP    172.16.10.44:62168     85.210.193.152:https   TIME_WAIT
  TCP    172.16.10.44:62169     85.210.193.152:https   TIME_WAIT
  TCP    172.16.10.44:62170     85.210.193.152:https   TIME_WAIT
  TCP    172.16.10.44:62171     85.210.193.152:https   TIME_WAIT
  TCP    172.16.10.44:62176     150.171.110.84:https   ESTABLISHED
  TCP    172.16.10.44:62177     a23-34-81-17:https     ESTABLISHED
  TCP    172.16.10.44:62178     123:http               ESTABLISHED
```

```
C:\Users\HDC0222053>nslookup www.gmail.com
Server:  UnKnown
Address:  172.16.8.1

Non-authoritative answer:
Name:    www.gmail.com
Addresses:  2404:6800:4007:835::2005
          142.251.222.197
```

```
C:\Windows\system32>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.

C:\Windows\system32>
```

```
C:\Windows\system32>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

```
C:\Windows\system32>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f           Clears the routing tables of all gateway entries.  If this is
               used in conjunction with one of the commands, the tables are
               cleared prior to running the command.

  -p           When used with the ADD command, makes a route persistent across
               boots of the system. By default, routes are not preserved
               when the system is restarted. Ignored for all other commands,
               which always affect the appropriate persistent routes.

  -4           Force using IPv4.

  -6           Force using IPv6.
```

Linux

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1. Importing from JSON

Linux Networking Commands:

1. ip - This is one of the basic commands every admin will need in daily work. It will show the address info., manipulate routing, plus display network various devices.

* Options: ip address show, ip address add, ip address del

2. ipconfig - The ipconfig command was/is a staple in many sysadmins' tool belt for configuring and troubleshooting networks.

3. mtr - It is a program (Matt's traceroute) with a command-line interface that serves as a network diagnostic and troubleshooting tool.

Options: mtr google.com, mtr -g, mtr -b, mtr -c.

4. tcpdump - The tcpdump command is useful for capturing and displaying packets.

5. ping - It is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP).

```
devyesh@localhost: $ ip -d address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 allmulti 0 minmtu 0 maxmtu 0 numtxqueues 1 numrxqueues 1 gso_max_size 5
so_max_size 524280 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether d6:11:89:28:98:2a brd ff:ff:ff:ff:ff:ff promiscuity 0 allmulti 0 minmtu 68 maxmtu 1500 numtxqueues 1 numrxqueues 1 gso_max_size
tso_max_size 65536 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536 parentbus virtio parentdev virtio0
    altname enxd6118928982a
    inet 192.168.64.4/24 brd 192.168.64.255 scope global dynamic noprefixroute enp0s1
       valid_lft 3469sec preferred_lft 3469sec
    inet6 fdd6:f977:504c:340d:d411:89ff:fe28:982a/64 scope global dynamic noprefixroute
       valid_lft 2591983sec preferred_lft 604783sec
    inet6 fe80::d411:89ff:fe28:982a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

```
devyesh@localhost:~$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether d6:11:89:28:98:2a brd ff:ff:ff:ff:ff:ff
    altname enxd6118928982a
    inet 192.168.64.4/24 brd 192.168.64.255 scope global dynamic noprefixroute enp0s1
       valid_lft 3290sec preferred_lft 3290sec
    inet6 fdd6:f977:504c:340d:d411:89ff:fe28:982a/64 scope global dynamic noprefixroute
       valid_lft 2591915sec preferred_lft 604715sec
    inet6 fe80::d411:89ff:fe28:982a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

```
devyesh@localhost:~$ ifconfig
enp0s1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.64.4  netmask 255.255.255.0  broadcast 192.168.64.255
        inet6 fdd6:f977:504c:340d:d411:89ff:fe28:982a  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::d411:89ff:fe28:982a  prefixlen 64  scopeid 0x20<link>
        ether d6:11:89:28:98:2a  txqueuelen 1000  (Ethernet)
        RX packets 2523  bytes 3209930 (3.0 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2312  bytes 206593 (201.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 18  bytes 2112 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 18  bytes 2112 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Host	Packets Loss%	Snt	Pings Last	Avg	Best	Wrst	StDev
1. _gateway	0.0%	93	0.5	0.5	0.3	0.8	0.1
2. 192.168.135.65	0.0%	93	16.3	10.7	3.2	133.6	14.0
3. **192.168.72.131**	80.4%	93	63.8	42.5	21.6	66.1	15.4
4. 192.168.116.5	0.0%	93	61.9	40.1	23.5	162.1	18.1
5. 192.168.116.49	73.9%	93	39.0	36.6	24.6	53.0	10.2
6. **(waiting for reply)**							
7. nsg-corporate-81.102.187.122.airtel.in	0.0%	93	59.1	41.1	22.5	65.9	11.9
8. **116.119.68.247**	27.2%	93	42.2	47.2	32.8	82.7	12.8
9. **142.250.169.206**	66.3%	92	44.5	44.6	32.2	68.6	10.5
0. 64.233.174.17	0.0%	92	50.6	45.8	30.9	72.4	9.5
1. 142.250.236.157	0.0%	92	42.9	43.3	31.3	67.4	8.8
2. lcmaaa-aq-in-f14.1e100.net	0.0%	92	45.5	43.8	30.8	65.9	9.6

```
devyesh@localhost:~$ ping google.com
PING google.com (142.251.43.142) 56(84) bytes of data.
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=1 ttl=113 time=74.2 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=2 ttl=113 time=73.3 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=3 ttl=113 time=65.5 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=4 ttl=113 time=61.5 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=5 ttl=113 time=58.3 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=6 ttl=113 time=63.1 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=7 ttl=113 time=63.7 ms
```

1. Importing from JSON

Result:

The various networking commands used for Linux & Windows for networking configuration and diagnostics has been studied.